A THT Report on

"A MAN WALKING IN THE RAIN"

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CERTIFICATE

This is to certify that the project report entitled "A Man Walking in the Rain" is submitted for partial fulfillment of Bachelor Degree in Computer Science And Engi neering as per requirement of Punyashlok Ahilyadevi Holkar Solapur University, Solapur for the academic year 2023-2024.

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Abstract

The "A Man Walking in the Rain" computer graphics mini-project aims to simulate a visually immersive scene of a man walking in the rain. This project combines principles of computer graphics and animation to create a realistic and dynamic environment. The main components include modeling a 2D character, animating the walking motion, simulating raindrops, and rendering the final scene.

Key Components:

2D Character Modeling: Design and model a 2D character representing a man. Consider details such as body proportions, features to achieve a realistic appearance.

Raindrop Simulation: Develop a raindrop simulation system to create a convincing rain effect. This involves generating and animating raindrops falling from the sky, taking into account factors like gravity and wind.

Environment Design: Construct a 2D environment that complements the rain scene. This may include a textured ground, and appropriate lighting to enhance realism.

Expected Outcome:

The final output of the mini-project will be a visually appealing and computer-generated scene where a realistically modeled man walks in a simulated rain environment. This project not only demonstrates proficiency in computer graphics but also provides an engaging experience for users with the rainy scene.

Keywords: Computer Graphics, 2D Modeling, Animation, Rain Simulation, Rendering, Environment.

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Chapter 1

Introduction

The "A Man Walking in the Rain" computer graphics project offers a captivating exploration into the real of virtual environments, animation, and simulation. In this endeavor, we delve into the artistry of crafting a visually compelling scene where a 2D-rendered man walks in a realistic rainstorm. The project includes of character modeling, animation, and environmental simulation to bring to life the immersive experience of strolling through a rain-soaked landscape.

Objectives:

Character Representation: Model a 2D character with meticulous attention to detail, capturing the nuances of human anatomy and movement.

Raindrop Simulation: Develop a sophisticated raindrop simulation system, considering physics principles to emulate the behavior of rain in a convincing manner.

Through "A Man Walking in the Rain," we aim to evoke emotions, spark curiosity, and highlight the limitless possibilities that emerge at the intersection of art and technology in the realism of computer graphics.

Chapter 2

Methodology

The "A Man Walking in the Rain" computer graphics project employs a comprehensive methodology that integrates various techniques in 2D modeling, animation, and simulation.

Conceptualization: Define the project's objectives, including creating a scene of a man walking in the rain.
Raindrop Simulation: Implement a raindrop simulation system to generate and animate raindrops falling from the sky.Consider physics principles such as gravity and wind to emulate realistic rain behavior.
Testing and Iteration: Conduct thorough testing to identify and address any issues related to character animation, rain simulation.
Optimization: Optimize the project for performance, ensuring smooth rendering and interaction even in resource-constrained environments.

This methodology ensures a systematic and holistic approach to creating a visually immersive and interactive computer graphics project featuring a man walking in the rain.

Chapter 3

Implementation

```
#include<conio.h>
#include<stdio.h>
#include<graphics.h>
void displayMan(int x,int y)
{
circle(x,y,10);
line(x,y+10,x,y+30);
line(x,y+30,x-20,y+40);
line(x,y+30,x+20,y+40);
line(x+20,y+40,x+30,y+30);
line(x,y+30,x,y+70);
line(x+30,y+30,x+30,y-90);
pieslice(x+30,y-30,0,180,55);
}
```

Figure 1: Displaying of Man.

```
void drawCloud(int z,int y)
{
int r=50;
pieslice(z+300,y,0,360,r);
pieslice(z,y,0,360,r);
pieslice(z+350,y,0,360,r);
pieslice(z+50,y,0,360,r);
pieslice(z+400,y,0,360,r);
pieslice(z+100,y,0,360,r);
}
```

Figure 2: Displaying of Cloud.

```
void main()
{
int gd=DETECT, gm,i,d=0,x=50,y=340,z=50,shouldMove=1;
int rx,ry;
initgraph(&gd,&gm,"C:\\TURBOC3\\BGI");

while(!kbhit())
{
cleardevice();
displayMan(x,340);

drawCloud(z,60);
line(0,430,639,430);
for(i=0;i<500;i++)
{
rx=rand()%639;
ry=rand()%439;
if(rx>=(x-40)&&rx<=(x+110))
  if(ry>=(y-50)&&ry<=479)
  continue;
  line(rx-10,ry+10,rx,ry);
```

Figure 3: Logic Part.

```
if(shouldMove)
{
if(d<20)
d+=4;
else
shouldMove=0;
line(x,y+70,x-d,y+90);
line(x,y+70,x+d,y+90);
}
else
{
if(d>0)
d-=4;
else
shouldMove=1;
line(x,y+70,x-d,y+90);
line(x,y+70,x+d,y+90);
}
delay(200);
x=(x+10)%639;
}
```

Figure 4: Logic part

Chapter 4

Result/Outputs

Figure 5: Output

Chapter 6

CONCLUSION

CONCLUSION :

In concluding the "A Man Walking in the Rain" computer graphics project, we have successfully navigated the intersection of art and technology to deliver a visually compelling and best experience. The synthesis of 2D modeling, animation, and rain simulation techniques has resulted in a scene that captures the realism of a man walking in the rain.

While the project stands as a statement to the capabilities of computer graphics in creating captivating virtual scenarios, there are avenues for future enhancement and exploration.